PRESENTATION for FEB 2026

Sections:

1. **Data Origin and Cohort Definitions**
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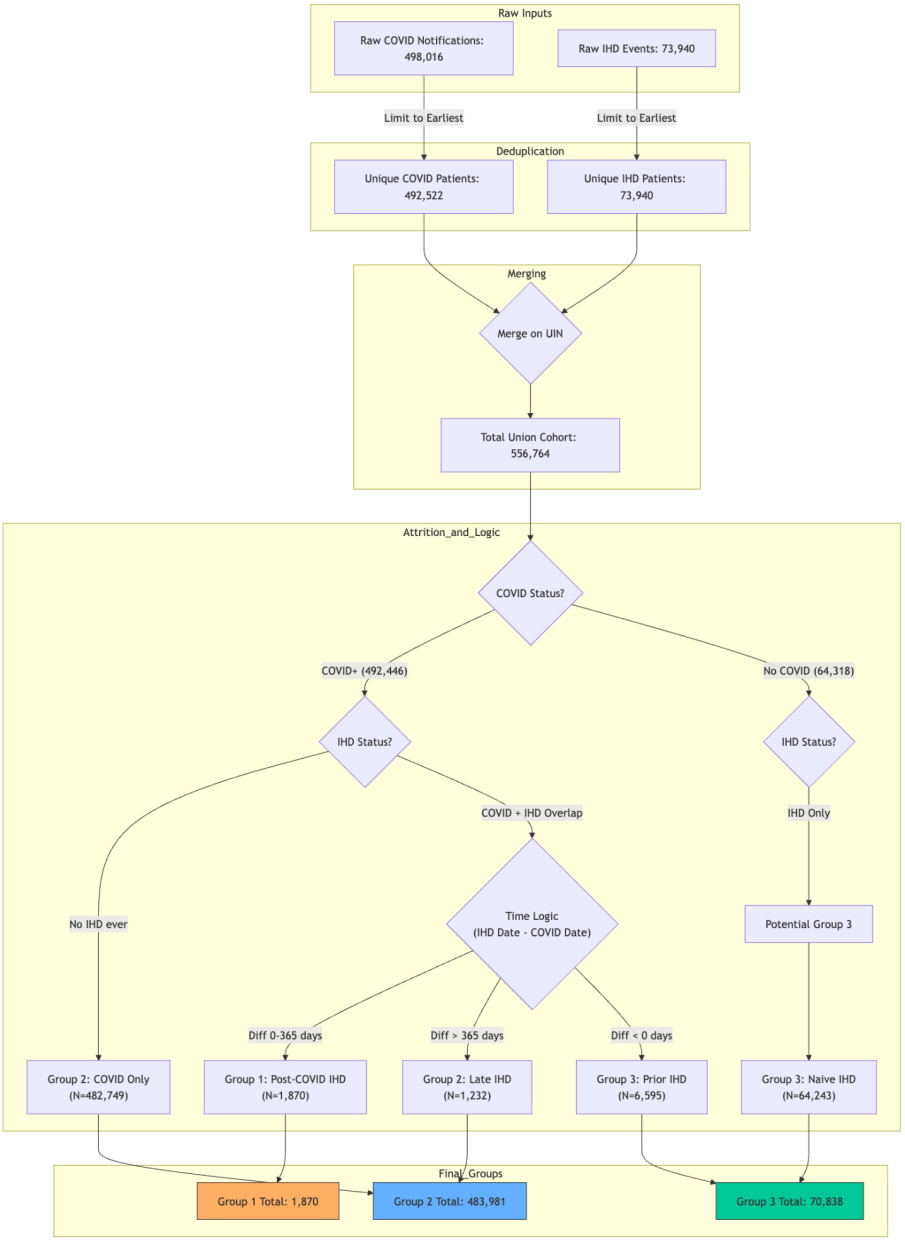
Background
While the acute respiratory manifestations of COVID-19 are well-documented, the long-term cardiovascular sequelae—specifically the incidence of Ischemic Heart Disease (IHD)—remain an area of critical concern. As the pandemic transitions into an endemic phase, understanding the burden of new-onset IHD in recovered patients is essential for guiding public health screening and resource allocation.

Objective
This study aimed to quantify the excess risk of incident IHD in patients with a prior confirmed COVID-19 infection compared to the general population. We sought to determine the Age-Standardized Incidence Rate (ASIR) of IHD in the post-COVID cohort and calculate the Standardized Incidence Ratio (SIR) to provide a direct risk comparison against the national background rate (derived from Singapore Census 2020).

Key Findings
Our analysis of the national COVID-19 cohort (484,896 person-years of follow-up) revealed the following:

Significantly Elevated Risk: Individuals with prior COVID-19 infection had an 85% higher risk of developing IHD compared to the general age-matched population (SIR 1.85; 95% CI: 1.77–1.94).
Incidence Burden: The Age-Standardized Incidence Rate (ASIR) for the post-COVID cohort was 463.57 per 100,000 person-years, highlighting a substantial absolute burden of disease in this group.Good

1. Data Origin



Data is taken from Mediclaims and Dengue Notification Datasets

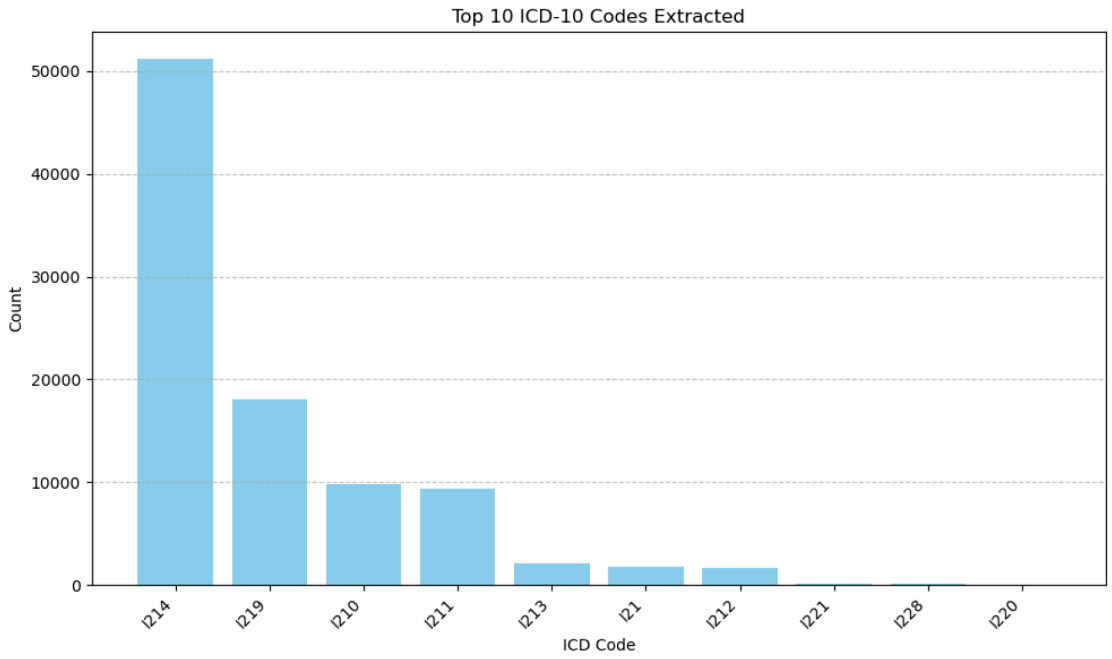
1.1 Cohort Definitions

1. COVID-19 Cohort (Exposed): Patients with a confirmed COVID-19 infection who subsequently developed IHD (Group 1) or remained IHD-free (Group 2). Person-years at risk were calculated from the date of infection until IHD diagnosis, death, or 365 days of follow-up.

2. Non-COVID Reference Population: Derived by subtracting the COVID-19 cohort counts from the Singapore Census 2020 population estimates per age group.

3. Naive IHD Cases (Group 3): Patients with incident IHD and no prior COVID-19 infection within 365 days. These were used to calculate the background incidence rates in the Non-COVID population.

1.2 ICD Codes used:

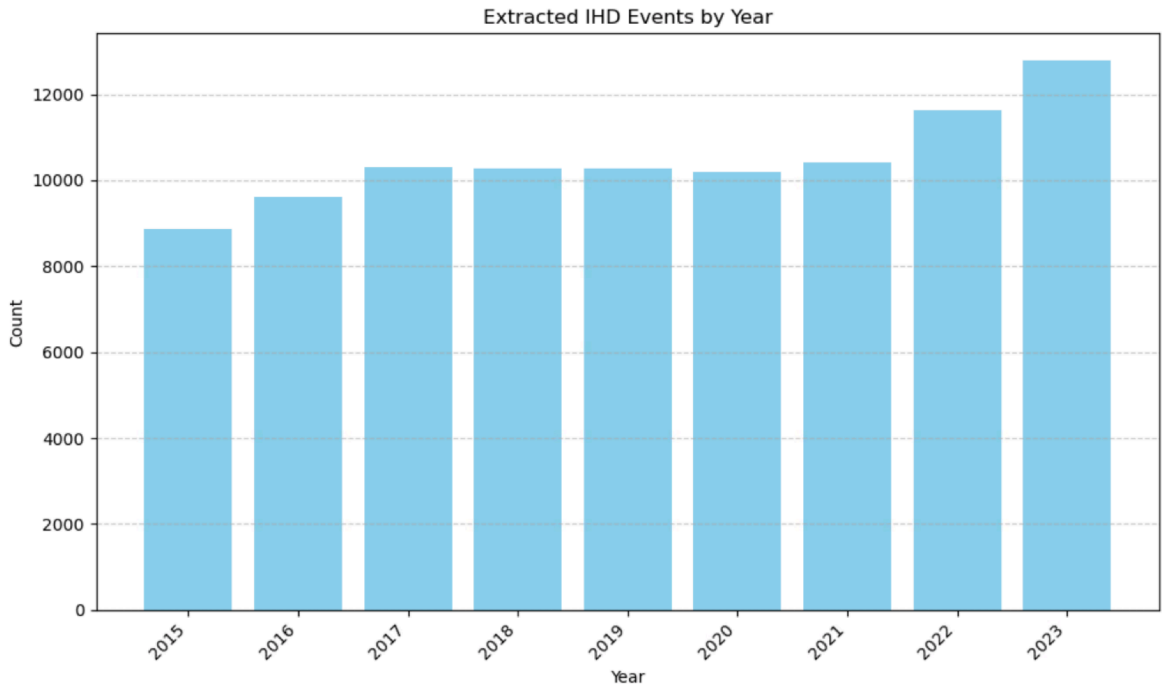


2. Table 1 Statistics

2.1 Statistics of 3 groups

Table 1: Baseline Characteristics of Study Population by Group			
Characteristic	**Group 1 (n=1,891)**	**Group 2 (n=484,501)**	**Group 3 (n=72,222)**
Demographics			
Age, years (mean ± SD)	71.5 ± 14.5	41.5 ± 22.2	69.0 ± 13.3
Male, %	65.8	59.8	54.1
Comorbidities (%)			
CKD	31.0	4.2	21.0
Diabetes	50.0	11.0	47.1
Heart Failure	21.9	1.1	20.5
Hyperlipidemia	63.5	16.2	60.6
Hypertension	75.2	17.6	69.9
Obesity	0.2	0.3	0.3
Prior IHD	100.0	4.2	100.0
Stroke	18.6	3.2	13.3

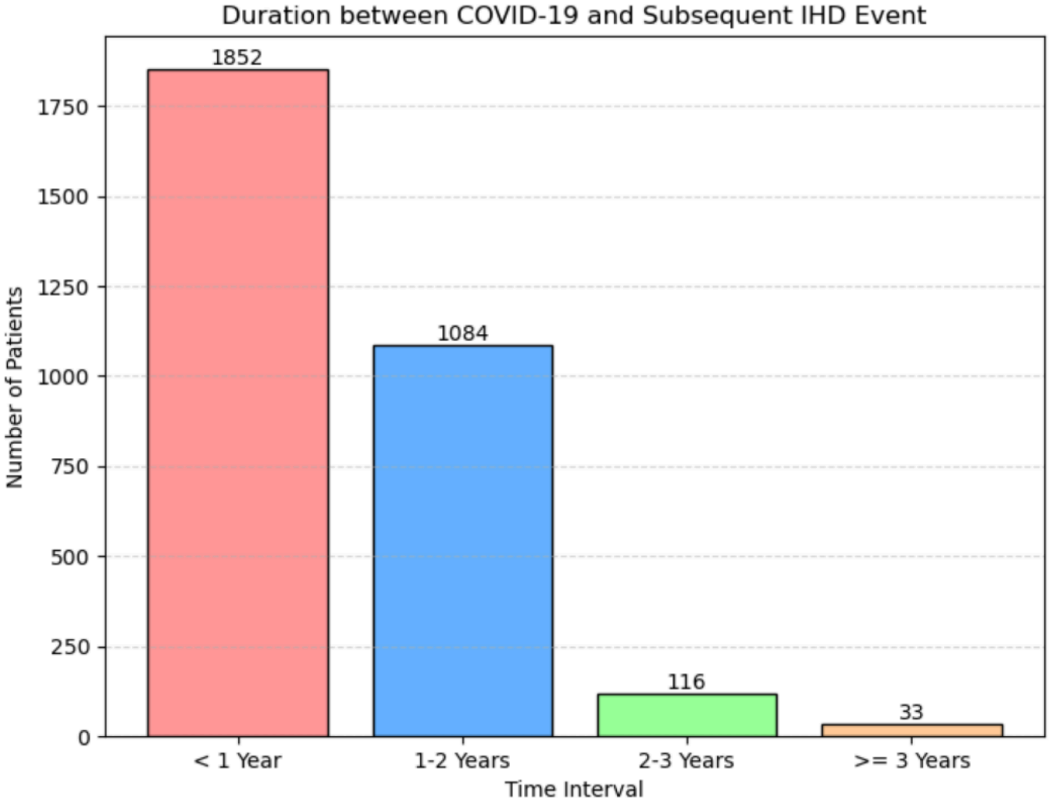
2.2 IHD Events per year



The IHD events by year in this table mirrors that of the SMIR report roughly, each year drops off about 1000 cases of AMI though.

Table 5.1.1: Incidence numb

Year of onset	Number
2012	9124
2013	9532
2014	9835
2015	10132
2016	10815
2017	11949
2018	11983
2019	12651
2020	11680


Duration between COVID-19 and Subsequent IHD Event
Number of Patients
Time Interval
1852
1084
116
33
0
250
500
750
1000
1250
1500
1750
< 1 Year
1-2 Years
2-3 Years
>= 3 Years

3. Methodology

Standardized Incidence Ratio (SIR):

Calculated to quantify the **relative risk of IHD in the COVID-19 cohort compared to the general population**.

The expected number of IHD cases in the COVID-19 cohort was estimated by applying the age-specific incidence rates of IHD of the Non-COVID reference population to the person-years accumulated by the COVID-19 cohort.

$$SIR = \frac{\text{Observed Events in COVID Cohort}}{\sum(\text{Person-Years}_{COVID,i} \times \text{Rate}_{Non-COVID,i})}$$

Formula for standardized incidence rate (SIR) - Byar's approximation was used for calculating 95% Confidence Intervals

Standardized Incidence Ratio (SIR):

Calculated using a direct method to allow for comparison with other registries (e.g., SMIR).

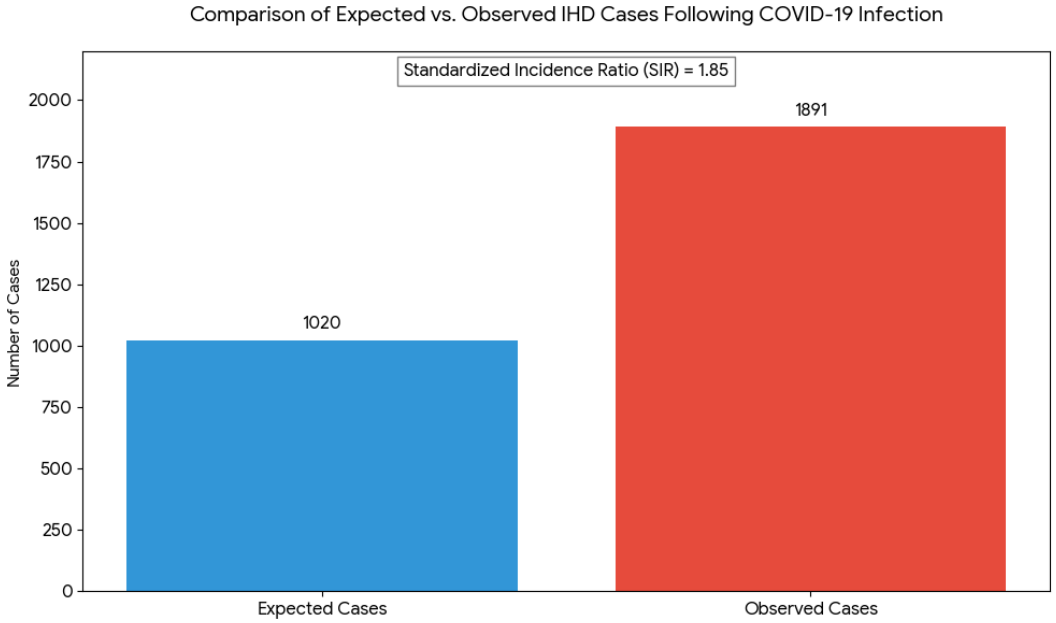
The age-specific incidence rates of the COVID-19 cohort were weighted according to the Singapore Census 2020 age distribution.

$$ASIR = \sum(\text{Rate}_{COVID,i} \times \text{Weight}_{Census,i})$$

Formula for standardized incidence rate (SIR) - Method replicates

4. Results

4.1 Does having COVID increase IHD Risk?

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[STEP 2] Standardized Incidence Ratio (SIR)
Total Observed (G1): 1891
Total Expected:      1020.16
SIR Calculation:     1891 / 1020.16
SIR Result:          1.8536 (95% CI: 1.7701 - 1.9372)
interpretation: COVID-19 group has 1.85x higher risk of IHD than general population.
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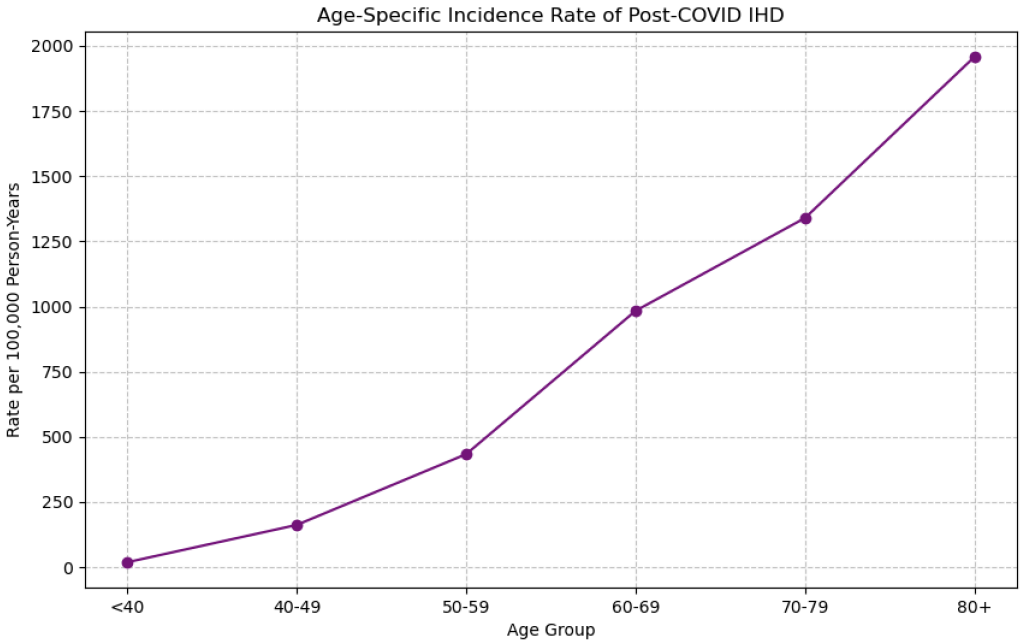
COVID-19 **significantly increases** the risk of IHD. It does *not* reduce it. The Standardized Incidence Ratio (SIR) is **1.85** (95% CI: 1.77 – 1.94).

- Individuals with a prior COVID-19 infection are **85% more likely** to develop IHD in the subsequent year compared to the general population of the same age and sex.
- We observed **1,891** cases in the COVID cohort, whereas only **1,020** were expected based on national background rates.

4.2 Age-Specific Burden Findings

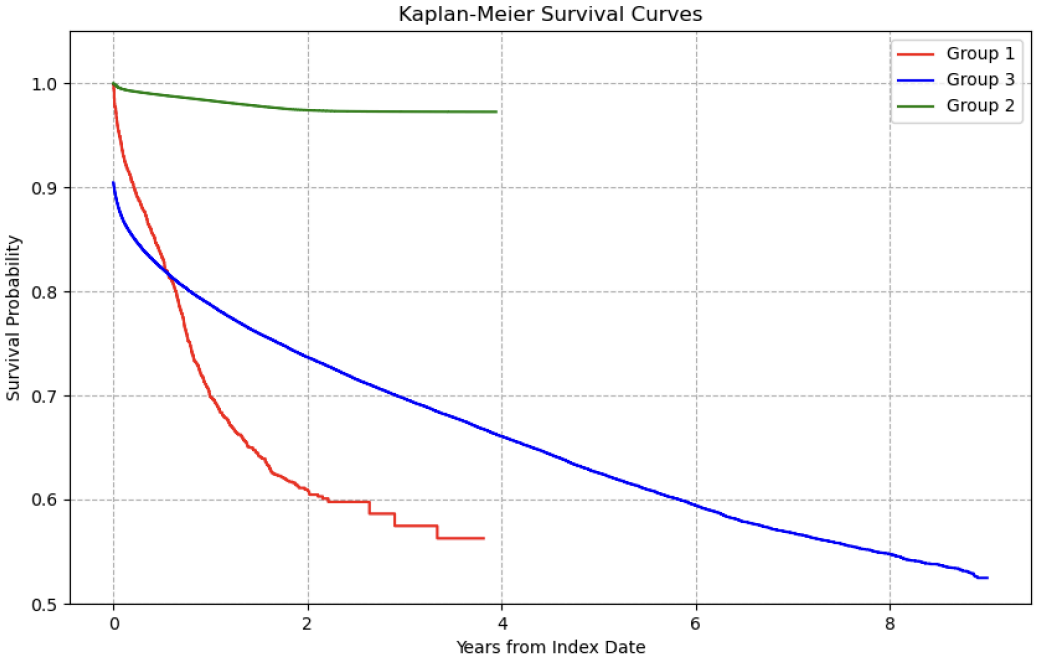
Findings from ASIR & Age Stratification



- **Absolute Burden**: The absolute rate of IHD increases exponentially with age, peaking in the >80 group (1,958 per 100k PY).
- **Relative Impact**:
    - While older adults have the highest number of cases, the *biological signal* of COVID-associated IHD is present even in younger groups (<40 age group showed a rate of 18.6 vs Reference ~10.9).
    - The **Age-Standardized Incidence Rate (ASIR)** is **463.57 per 100,000**, confirming that this burden is substantial even after adjusting for Singapore's aging population structure.

4.3: Survival Analysis (Kaplan-Meier Curves)

Interpretation of the "Double Hit" Effect



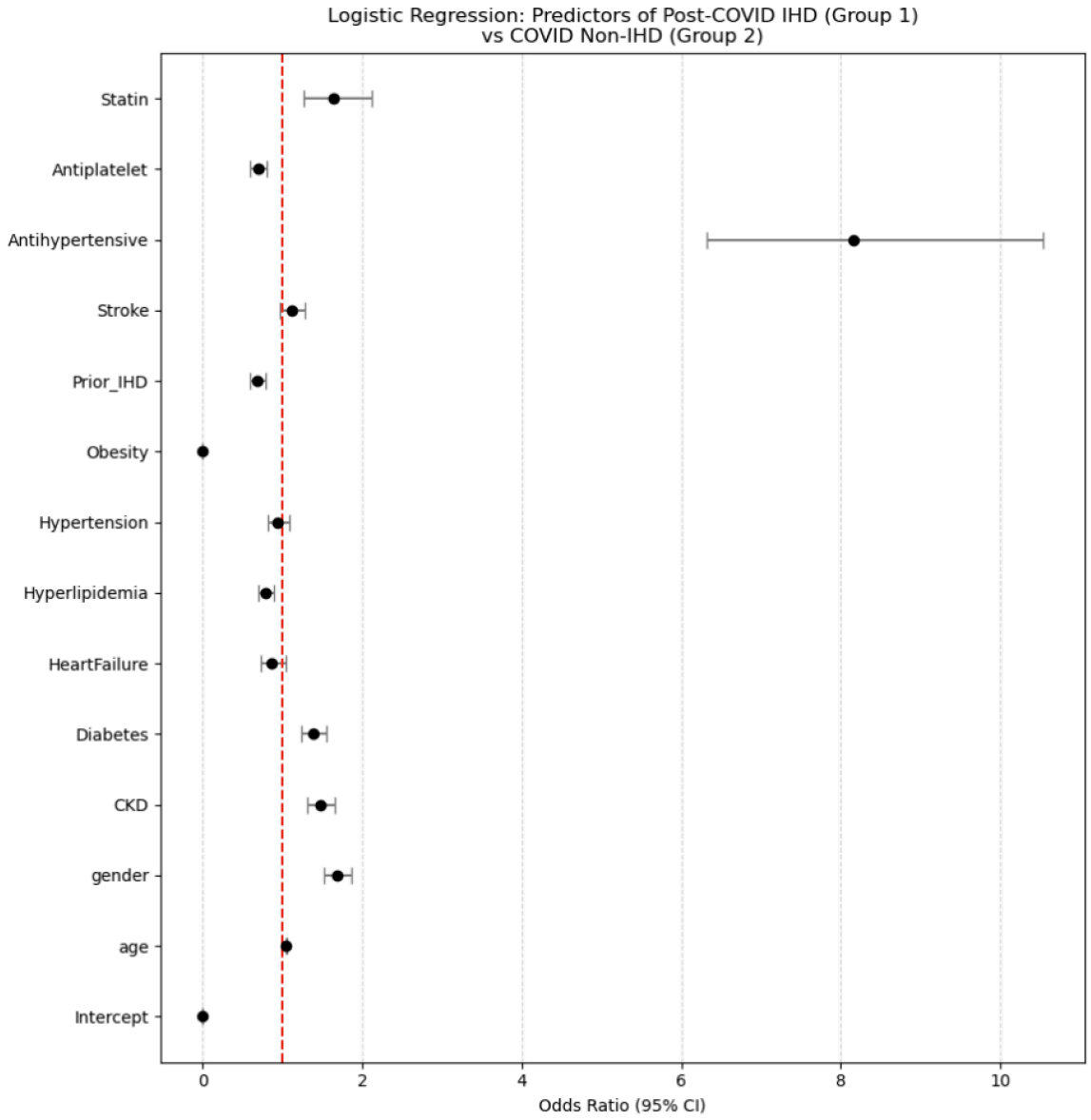
- **Red Line (Group 1: Post-COVID IHD)**: Shows the steepest decline, dropping to ~55% survival by Year 4.
- **Blue Line (Group 3: Naive IHD)**: Shows a moderate decline, maintaining ~65% survival by Year 4.
- **Green Line (Group 2: COVID Non-IHD)**: Remains highly stable (>95% survival), serving as a control.

Insights:

- Patients who develop IHD *after* COVID-19 (Red) have significantly **higher mortality** than those with IHD alone (Blue).
- This suggests a **synergistic effect**: The physiological stress of COVID-19 combined with an ischemic event leads to poorer outcomes than ischemia alone. The "Red vs Blue" gap represents the *excess mortality* attributable to the recent infection context.

4.4 Predictors of Post-COVID IHD

Logistic Regression (Forest Plot)



- **Strongest Risk Factor**: **Antihypertensive Medication Use** (OR ~8.0).
    - *Interpretation*: This is a powerful proxy for underlying cardiovascular vulnerability. Patients with established hypertension requiring medication are vastly more susceptible to triggering an IHD event during the post-acute COVID window.
- **Confounding by Indication**:
    - **Statin** (OR ~1.2) and **Antiplatelets** (OR ~0.7-0.8): The fact that Statin use predicts IHD suggests these patients were *already identified as high-risk* prior to infection. They didn't get IHD *because* of statins; they took statins because they were already prone to IHD ("Sick User Effect").
- **Protective Factors?**:
    - **Obesity** (OR < 1.0): This "Obesity Paradox" is common in observational administrative data—obese patients might be younger or receiving more aggressive monitoring, or this reflects survival bias (frail non-obese elderly dying faster).
- **Demographics**: Age and Male Gender (OR > 1) remain classic, unchangeable risk factors.

The patients most likely to suffer Post-COVID IHD are those with **pre-existing metabolic and vascular burden** (Hypertension, Prior IHD history), essentially "tipping them over the edge."